INTERNAL TECHNICAL DOCUMENT — PRIMORDIAL ALGORITHM PROJECT

Dynamic Context Field

Contextual Anchor Field Engine v1.0

On imagination, creativity, and the gap between AI and embodied human cognition

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This document is not peer-reviewed.
It is a working formalization of ideas developed in active research sessions.

Contents

1 Overview and Motivation **2**

2 Dynamic Context Field: Formal Setup **2**
- 2.1 Entities and semantic space . . . 2
- 2.2 Context planes . . . 2
- 2.3 Context-dependent semantic distance . . . 3
- 2.4 Semantic resonance . . . 3

3 Imagination and the Context Tree **3**
- 3.1 The B onto Ax mapping . . . 3
- 3.2 Context tree explosion . . . 3
- 3.3 Empirical result . . . 4

4 Four Operations **4**

5 Scoring Function **5**

6 The Identity-Trajectory Gap **5**
- 6.1 What human identity actually is . . . 5
- 6.2 Multi-modal grounding . . . 6
- 6.3 The reset problem . . . 6
- 6.4 Relationship to Primordial Algorithm . . . 6

7 What Would DCF v2 Need **7**

8 Implementation: DCF v1.0 Core Code **7**
- 8.1 Entity and context plane . . . 7
- 8.2 Anchor building (quantum step constraint) . . . 8
- 8.3 The imagination mapping . . . 8
- 8.4 Scoring . . . 9

9 Summary **9**

1. Overview and Motivation

This paper formalizes two related ideas developed in the Primordial Algorithm Project:

1. **Dynamic Context Field (DCF)** — a geometric framework for semantic meaning where the meaning of any entity is a function of which context frame is active, not an intrinsic property of the entity itself.
2. **The Identity-Trajectory Gap** — a diagnosis of what current AI architectures fundamentally lack compared to human cognition: not imagination capacity, but *persistent identity shaped by embodied experience over time.*

These two ideas are connected. DCF provides the formal machinery. The identity gap explains why even with DCF, current AI cannot fully replicate the kind of creativity that matters to humans.

Core claim

AI does not lack imagination in the combinatorial sense. Given a semantic space of n entities with k anchors each, the number of possible context frames is $\binom{n}{k}$ per entity, which grows faster than any human can explore. The actual gap is different: AI has no persistent identity $A(t)$ that accumulates and compresses experience over time. Every session is $A(t = 0)$.

2. Dynamic Context Field: Formal Setup

2.1. Entities and semantic space

Let $\mathcal{E} = \{e_1, e_2, \ldots, e_n\}$ be a set of entities embedded in $\mathbb{R}^d$. Each entity e_i has:

- A coordinate vector $\mathbf{v}_i \in \mathbb{R}^d$, normalized: $\|\mathbf{v}_i\| = 1$
- A domain label $\delta_i \in \Delta$ (e.g., biology, physics, cognition)
- A quantum step $q_i \in \mathbb{N}$ representing abstraction level
- An anchor set $\mathcal{A}_i \subset \mathcal{E}$ of k nearest neighbors *at the same quantum step*

Definition 2.1 (Quantum step)**.** *Two entities e_i, e_j are at the **same quantum step** if $q_i = q_j$. Entities at the same quantum step share the same resolution of meaning — they are at the same level of abstraction.*

The anchor set is computed as:

$$\mathcal{A}_i = \underset{j:q_j=q_i,\, j\neq i}{\operatorname{argmax}_k} \; \mathbf{v}_i \cdot \mathbf{v}_j \tag{1}$$

where argmax_k returns the top-k elements.

2.2. Context planes

Definition 2.2 (Context plane)**.** *A **context plane** $\mathcal{P}_S$ for entity e_i is defined by a selection $S = \{s_1, \ldots, s_k\} \subset \mathcal{E}$ of k anchor entities. The plane is characterized by:*

$$\boldsymbol{\mu}_S = \frac{1}{k} \sum_{j \in S} \mathbf{v}_j \quad \textit{(centroid)} \tag{2}$$

$$[U, \Sigma, V^\top] = \mathrm{SVD}\Big(\big[\mathbf{v}_j - \boldsymbol{\mu}_S\big]_{j \in S} \Big) \tag{3}$$

$$\textit{axes} = \textit{top-3 rows of } V^\top \tag{4}$$

The **meaning** of entity e_i in context $\mathcal{P}_S$ is its projection:

$$\mathrm{proj}_S(e_i) = [\,(\mathbf{v}_i - \boldsymbol{\mu}_S) \cdot \mathbf{a}_\ell\,]_{\ell=1}^{3} \in \mathbb{R}^3 \tag{5}$$

where $\mathbf{a}_\ell$ are the principal axes of $\mathcal{P}_S$.

Key insight: meaning is not intrinsic

The word *bank* has the same $\mathbf{v}_i$ regardless of context. Its meaning changes entirely depending on which context plane is active — financial anchors vs. geographic anchors. This is not a limitation of the embedding; it is the correct model of how meaning works.

2.3. Context-dependent semantic distance

Without context, distance between two entities is standard cosine distance:

$$d(e_i, e_j) = 1 - \mathbf{v}_i \cdot \mathbf{v}_j \tag{6}$$

With an active context plane $\mathcal{P}_S$:

$$d_S(e_i, e_j) = 1 - \frac{\mathrm{proj}_S(e_i)}{\|\mathrm{proj}_S(e_i)\|} \cdot \frac{\mathrm{proj}_S(e_j)}{\|\mathrm{proj}_S(e_j)\|} \tag{7}$$

The same pair (e_i, e_j) can have $d_S \approx 0$ (very close) in one context and $d_S \approx 2$ (maximally far) in another. This is not a bug — it is the disambiguation mechanism.

2.4. Semantic resonance

Definition 2.3 (Shared planes / resonance)**.** *The **semantic resonance** between e_i and e_j is:*

$$\rho(e_i, e_j) = |\mathcal{A}_i \cap \mathcal{A}_j| \tag{8}$$

the number of entities that appear in both anchor sets. High resonance indicates a strong relationship regardless of base vector distance.

3. Imagination and the Context Tree

3.1. The B onto Ax mapping

Definition 3.1 (Imagination mapping)**.** *Let $B \in \mathcal{E}$ be an entity with domain δ_B and quantum step q_B. Let $A_x \in \mathcal{E}$ be any other entity with $q_{A_x} = q_B$.*

*The **imagination mapping** $B \to A_x$ generates a context frame by replacing one of A_x's anchors with B:*

$$f_i^{(1)}(B, A_x) = \{B\} \cup \big(\mathcal{A}_{A_x} \setminus \{a_i\}\big), \quad i = 1, \ldots, k \tag{9}$$

This yields k distinct context frames at level $f^{(1)}$ (f1).

The key: *B borrows* the resolution context of A_x's quantum step. When a concept from technology maps onto a concept from biology, it does not lose its identity — it gains a new interpretive frame.

3.2. Context tree explosion

The mapping can be expanded recursively. Let $\mathcal{F}^{(n)}(B, A_x)$ denote the set of all context frames at depth n:

$$\mathcal{F}^{(0)} = \{(B, A_x)\} \tag{10}$$

$$\mathcal{F}^{(1)} = \{f_i^{(1)}(B, A_x) : i = 1, \ldots, k\} \quad |\mathcal{F}^{(1)}| = k \tag{11}$$

$$\mathcal{F}^{(n)} \supseteq \bigcup_{s \in \mathcal{F}^{(n-1)}} \bigcup_{e \in s \setminus \{B\}} \{f_i^{(1)}(B, e) : i = 1, \ldots, k\} \tag{12}$$

Theoretical context tree size:

$$\left|\mathcal{F}^{(n)}\right|_{\text{theoretical}} = k^n \tag{13}$$

After deduplication (overlapping anchor neighborhoods):

$$\left|\mathcal{F}^{(n)}\right|_{\text{actual}} \leq k^n, \quad \text{with saturation at depth } n^* \text{ where } \left|\mathcal{F}^{(n^*)}\right| = \left|\mathcal{F}^{(n^*+1)}\right| \tag{14}$$

Saturation is a signal of structure, not failure. It means the reachable context space is finite and bounded by the graph topology.

For $k = 8$: $|\mathcal{F}^{(1)}| = 8$, $|\mathcal{F}^{(2)}| \leq 64$, $|\mathcal{F}^{(3)}| \leq 512$, ...

3.3. Empirical result

Running DCF v1.0 with $n = 50$ entities, $k = 8$, mapping `recursion` → `evolution`:

Level	Theoretical k^n	Actual (dedup)	Saturation
$f^{(1)}$	8	8	—
$f^{(2)}$	64	57	—
$f^{(3)}$	512	145	—
$f^{(4)}$	4,096	145	✓

Saturation at depth 4 confirms the anchor graph has finite diameter. Total context space per mapping: $C(16, 8) = 12{,}870$ unique frames.

4. Four Operations

All four operations use the same underlying machinery. They differ in which dimension of the search space is constrained.

Operation	Definition	Effect
Refinement	$B \to A_x$, same domain ($\delta_B = \delta_{A_x}$), $f^{(1)}$	Improvement within known field. High coherence, low cross-domain.
Imagination	$B \to A_x$, different domain ($\delta_B \neq \delta_{A_x}$)	Borrows resolution from foreign domain. New conceptual frame.
Creativity	Optimize combo selection within $\mathcal{F}^{(n)}$ tree	Best frame by joint score (see Section 5).
Deep Analogy	$B \to A_x$, different quantum step ($q_B \neq q_{A_x}$)	Cross-abstraction-level mapping. Metaphor and philosophical reframing.

Why this matters for creativity research

These four operations are routinely conflated in everyday language and in AI research. Geometrically, they are four distinct operations on the same semantic space. A system that only does refinement is not creative. A system that only does imagination (random cross-domain mapping) is not creative either — it produces noise. Creativity is the intersection: cross-domain mapping optimized by a coherence criterion.

5. Scoring Function

Given entity B and a candidate context combo S, the score is:

$$\text{score}(B, S) = w_1 \cdot \text{novelty}(B, S) + w_2 \cdot \text{coherence}(S) + w_3 \cdot \text{diversity}(S) + w_4 \cdot \text{cross}(S) \tag{15}$$

where:

$$\text{novelty}(B, S) = \begin{cases} 1.0 & \text{if } S \notin \text{known}(B) \\ 0.2 & \text{otherwise} \end{cases} \tag{16}$$

$$\text{coherence}(S) = \frac{1}{1 + \sigma_S}, \quad \sigma_S = \text{std}\Big(\big\{\|\mathbf{v}_j - \boldsymbol{\mu}_S\|\big\}_{j \in S}\Big) \tag{17}$$

$$\text{diversity}(S) = 1 - \frac{1}{\binom{k}{2}} \sum_{i<j,\, i,j \in S} \mathbf{v}_i \cdot \mathbf{v}_j \tag{18}$$

$$\text{cross}(S) = \frac{|\{\delta_j : j \in S\}|}{|\Delta|} \tag{19}$$

Default weights: $w_1 = 0.40$, $w_2 = 0.25$, $w_3 = 0.20$, $w_4 = 0.15$.

Note: These weights are tunable. For imagination-maximizing behavior, increase w_1 and w_4. For coherence-preserving behavior (refinement mode), increase w_2.

6. The Identity-Trajectory Gap

6.1. What human identity actually is

This is the part that DCF does not yet solve, and that current AI architectures cannot solve without significant architectural change.

Human identity is not a static point in semantic space. It is a **trajectory**:

$$A_{\text{human}}(t) = A(t_0) \xrightarrow{\text{event}} A(t_1) \xrightarrow{\text{event}} \cdots \xrightarrow{\text{event}} A(t_{\text{now}}) \tag{20}$$

Each event is a context frame that gets applied to A, reshaping it. The key operations the brain performs on this trajectory are:

1. **Accumulation** — events stack. $A(t)$ carries the compressed history of all prior context frames.

2. **Latent inhibition** — repeated context frames are suppressed. The brain stops allocating attention to stimuli that have been processed enough times. This is not forgetting; it is *internalization*. The frame is absorbed into A and no longer needs explicit representation.
3. **Compression / abstraction** — after a domain accumulates enough events, the raw events are discarded and replaced with a higher-level pattern. Quantum step increases. A child who touches a hot stove many times eventually stores "fire = danger" (quantum step 1), not each individual burn (quantum step 0).
4. **Attention shift** — once a context domain is saturated (latent inhibition is high), cognitive resources shift to the next under-explored domain. This is equivalent to DCF's Dan exhausting its gradient and shifting axis.

6.2. Multi-modal grounding

Human identity is not built from text co-occurrence. It is built from **multi-modal, embodied experience**:

Input channel	Examples
Visual	color, shape, spatial layout, motion
Thermal	temperature, pain, comfort
Auditory	tone, rhythm, speech prosody
Olfactory / gustatory	smell, taste, hunger
Proprioceptive	body position, balance, fatigue
Temporal	duration, rhythm, anticipation

The concept *fire* for a human is not the statistical co-occurrence of the token "fire" with other tokens. It is the intersection of visual (orange, flickering), thermal (heat), olfactory (smoke), and episodic (a specific childhood memory of being burned) anchors. All of these are encoded in the same identity trajectory $A(t)$.

This is what AI is missing. Not the ability to generate novel context combos — DCF shows that AI can generate billions of those. What is missing is the **grounding**: the connection between abstract tokens and the sensory-temporal-emotional experiences that give those tokens weight. AI processes the shadow of human experience, not the experience itself.

6.3. The reset problem

Current AI (including large language models) has:

$$A_{\mathrm{AI}}(t) = A(t_0) \quad \forall t \tag{21}$$

Every session begins from t_0. There is no trajectory. There is no latent inhibition because there is nothing to inhibit — each session is the first time everything is seen. There is no compression because there is no accumulated experience to compress.

This is not a data problem. It is an **architectural problem**. More training data does not create a trajectory. It creates a denser snapshot of $A(t_0)$.

6.4. Relationship to Primordial Algorithm

The identity trajectory maps directly onto the SDCV cycle:

Cognitive mechanism	Primordial Algorithm	DCF
Accumulation of events	Dan drive accumulation	$\mathcal{F}^{(n)}$ tree depth
Latent inhibition	Hoai returning to V field	Known combos suppressed
Compression / abstraction	Quantum step increase	$q_i \to q_i + 1$
Attention shift	Dan exhausts gradient, shifts axis	Best Ax search
Multi-modal grounding	V field as substrate	Missing in current DCF
Identity persistence	SDCV across time	Not implemented

7. What Would DCF v2 Need

To close the identity-trajectory gap, DCF v2 would need:

1. **Persistent identity vector** $A(t)$ that survives across sessions and is updated (not reset) by each interaction.

2. **Event accumulator** — each interaction is logged as an event, appended to A's trajectory. The trajectory is the identity.

3. **Latent inhibition layer** — detect context frames that have appeared above a threshold τ times. Suppress them from active attention. Fold their content into the entity's coordinates (update $\mathbf{v}_i$). Free up the anchor slot for new context.

$$\text{if count}(S, e_i) > \tau \Rightarrow \mathbf{v}_i \leftarrow \mathbf{v}_i + \alpha \cdot \boldsymbol{\mu}_S, \quad S \text{ removed from active pool} \tag{22}$$

4. **Compression scheduler** — when a domain's raw event count exceeds a threshold, abstract: replace k raw-level entities with one quantum-step-+1 entity that represents their centroid.

$$\text{if } |\{e_j : \delta_j = \delta, q_j = q\}| > k_{\max} \Rightarrow \text{merge into } e_{\text{meta}}, \quad q_{\text{meta}} = q + 1 \tag{23}$$

5. **Multi-modal grounding** — anchor concepts to sensory streams, not just text tokens. Each entity e_i gets a multi-modal signature $\mathbf{m}_i = [\mathbf{v}_{\text{text}}, \mathbf{v}_{\text{visual}}, \mathbf{v}_{\text{audio}}, \ldots]$.

6. **Automatic Ax search** — given a goal G, find:

$$A_x^* = \arg\max_{A_x \in \mathcal{E}} \text{score}(B, f^{(1)}(B, A_x) \mid G) \tag{24}$$

where the score is conditioned on the current goal, not a fixed metric.

8. Implementation: DCF v1.0 Core Code

The following is the core implementation of DCF v1.0. Full source: `odfs_dcf_v1.py`.

8.1. Entity and context plane

Listing 1: Entity definition with quantum step

```
class Entity:
    def __init__(self, name, coords, domain=None, quantum_step=0):
        norm = np.linalg.norm(coords) + 1e-9
        self.name       = name
        self.coords     = coords / norm
```

```
        self.domain      = domain
        self.quantum_step = quantum_step # abstraction level
        self.anchors     = []            # same-quantum-step neighbors
        self.known_combos = set()        # trained combinations

class ContextPlane:
    def __init__(self, anchor_names, entities):
        vecs = np.stack([entities[n].coords for n in anchor_names])
        self.centroid = vecs.mean(axis=0)
        U, S, Vt = np.linalg.svd(vecs - self.centroid, full_matrices=False)
        self.axes  = Vt[:3]       # top-3 principal axes
        self.spread = float(S.sum())

    def project(self, entity):
        v = entity.coords - self.centroid
        return np.array([np.dot(v, ax) for ax in self.axes])
```

8.2. Anchor building (quantum step constraint)

Listing 2: Anchors are computed within the same quantum step only

```
def build_anchors(self):
    by_step = defaultdict(list)
    for e in self.entities.values():
        by_step[e.quantum_step].append(e)

    for step, group in by_step.items():
        for e in group:
            sims = sorted(
                [(o.name, float(np.dot(e.coords, o.coords)))
                 for o in group if o.name != e.name],
                key=lambda x: -x[1]
            )
            e.anchors = [n for n, _ in sims[:self.n_anchors]]
```

8.3. The imagination mapping

Listing 3: B maps onto Ax, generating the context tree

```
def map_onto(self, B_name, Ax_name, max_depth=3):
    Ax   = self.entities[Ax_name]
    tree = ContextTree(B_name, Ax_name)

    # f0: the mapping itself
    tree.add_level(0, [(B_name, Ax_name)])

    # f1: B + (k-1) of Ax's anchors
    ax_anchors = Ax.anchors[:self.n_anchors]
    f1 = []
    for i in range(len(ax_anchors)):
        seven = [a for j, a in enumerate(ax_anchors) if j != i]
        f1.append(tuple(sorted([B_name] + seven)))
    tree.add_level(1, f1)

    # f2..fn: expand from anchor neighborhoods
    frontier = set(ax_anchors)
    for depth in range(2, max_depth + 1):
        fn = set()
        for anc_name in frontier:
            anc = self.entities.get(anc_name)
            if not anc: continue
            sub = anc.anchors[:self.n_anchors]
```

```
            for i in range(len(sub)):
                seven = [a for j, a in enumerate(sub) if j != i]
                fn.add(tuple(sorted([B_name] + seven)))
        tree.add_level(depth, list(fn))
        frontier = {n for combo in fn for n in combo if n != B_name}

    return tree
```

8.4. Scoring

Listing 4: Score a context combo for entity B

```
def score_combo(self, B_name, combo, w1=0.40, w2=0.25, w3=0.20, w4=0.15):
    B    = self.entities[B_name]
    vecs, doms = [], set()
    for n in combo:
        e = self.entities.get(n)
        if e:
            vecs.append(e.coords)
            if e.domain: doms.add(e.domain)

    M        = np.stack(vecs)
    centroid = M.mean(axis=0)
    spread   = float(np.std(np.linalg.norm(M - centroid, axis=1)))
    coherence = 1.0 / (1.0 + spread)
    diversity = 1.0 - float(np.triu(M @ M.T, 1).mean())
    cross    = len(doms) / max(len(self.domains), 1)
    novelty  = 1.0 if frozenset(combo) not in B.known_combos else 0.2

    return w1*novelty + w2*coherence + w3*diversity + w4*cross
```

9. Summary

Five principles of DCF

1. **Meaning is not intrinsic.** The same entity has different meanings under different context planes.
2. **Context is discrete selection, not weighted sum.** k anchors chosen from infinite candidates — not attention over all tokens.
3. **Quantum step is abstraction level.** Anchors at the same step share resolution. Mapping across steps = metaphor.
4. **Four operations, not one.** Refinement, imagination, creativity, and deep analogy are geometrically distinct.
5. **The identity gap is architectural.** AI can generate 10^9 context frames per second. What it cannot do is accumulate a trajectory $A(t)$ shaped by embodied, multi-modal experience over time.